

Orthogonal projection and its technical applications

The drawing convention every engineer in the world uses — three views, and the object they determine.

LESSON 45–47

3 × 40 MIN

GEOMETRY 10, §18

IGCSE E14.1

LESSON CLOCK

14'

26'

30'

26'

30'

14'

Orthogonal versus general	14 min
The three views	26 min
Reconstructing a solid	30 min
The area formula	26 min
Practice	30 min
Homework	14 min

LEARNING OBJECTIVES

- Distinguish orthogonal from general parallel projection.
- Draw the plan, front and side elevations of a solid.
- Reconstruct a solid from its three views.
- Use the area formula for the orthogonal projection of a plane figure.

TEXTBOOK

National	pp. 211–228
Cambridge	Core & Extended, pp. 210–216
Workbook	Exercise 18.1

01

Explanation

Orthogonal versus general

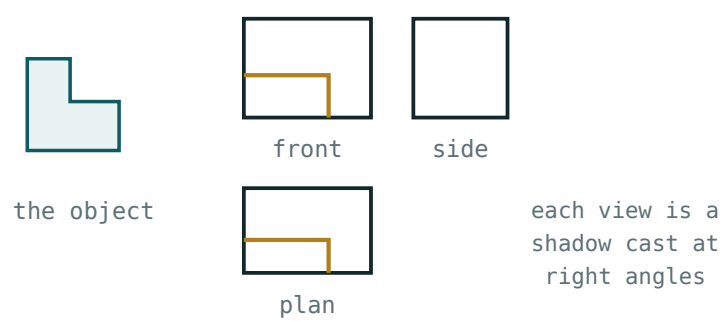
DEFINITION

Orthogonal projection is parallel projection in which the direction of projection is **perpendicular** to the plane of projection. Every point drops straight down onto the plane.

It is the special case that engineers use, because it preserves one thing a general parallel projection does not: **lengths parallel to the plane of projection are unchanged**. A drawing in orthogonal projection can therefore be measured — in that direction.

The projection of a point is the foot of the perpendicular; the projection of a segment is the segment between the feet; the projection of an oblique is what Lesson 35–37 called its projection.

The three views



three orthogonal projections determine the solid

*The plan from above, the front elevation from the front,
the side elevation from the side.*

VIEW	LOOKING	SHOWS
plan	from above	length and width
front elevation	from the front	length and height
side elevation	from the side	width and height

THE CONVENTIONS

Views are aligned: the plan sits directly below the front elevation, and the side elevation directly beside it, so that matching dimensions line up. Hidden edges are drawn dashed. Every view is drawn to the same scale.

Reconstructing a solid

Three views normally determine the solid — but not always. Two different solids can share all three views if the ambiguity is hidden inside.

PLAN	FRONT	SIDE	SOLID
circle	rectangle	rectangle	cylinder, upright
circle	triangle	triangle	cone
circle	circle	circle	sphere
square	square	square	cube
square	triangle	triangle	square-based pyramid

Reading a set of views is the everyday skill of a workshop; drawing them is the everyday skill of a design office. Both are the same theorem applied twice.

The area of a projection

THEOREM

If a plane figure of area S lies in a plane making a dihedral angle φ with the plane of projection, its orthogonal projection has area

$$S' = S \cdot \cos \varphi$$

Because lengths along the edge are unchanged and lengths perpendicular to it are multiplied by $\cos \varphi$, the area is scaled once.

Example. A square of side 10 lies in a plane at 60° to the horizontal. Its shadow on the horizontal has area $100 \cos 60^\circ = 50$.

THE ANGLE IS THE DIHEDRAL ANGLE

Not the angle of a line, and not the angle to the normal. If the figure's plane is at φ to the projection plane, the factor is $\cos \varphi$ — and $\varphi = 90^\circ$ collapses the figure to a segment, as it should.

02 Worked examples

EXAMPLE 1 A solid has a circular plan and rectangular front and side elevations. Name it.

- ① Circular from above.
- ② Rectangular from the front and side.
- ③ Constant circular cross-section.

ANSWER

An upright cylinder

EXAMPLE 2 A triangle of area 48 cm^2 lies in a plane at 30° to the horizontal. Find the area of its plan.

- ① $S' = S \cos \varphi$
- ② $= 48 \cos 30^\circ$
- ③ $= 48 \times 0.866 \approx 41.6$

ANSWER

$$24\sqrt{3} \approx 41.6 \text{ cm}^2$$

EXAMPLE 3 Cube of edge 6. Find the area of the orthogonal projection of the section ABC_1D_1 onto the base.

① The section is a rectangle $6 \times 6\sqrt{2}$, area $36\sqrt{2}$.

② It makes 45° with the base.

From Lesson 40–42.

③ $36\sqrt{2} \times \cos 45^\circ = 36$

The base square itself.

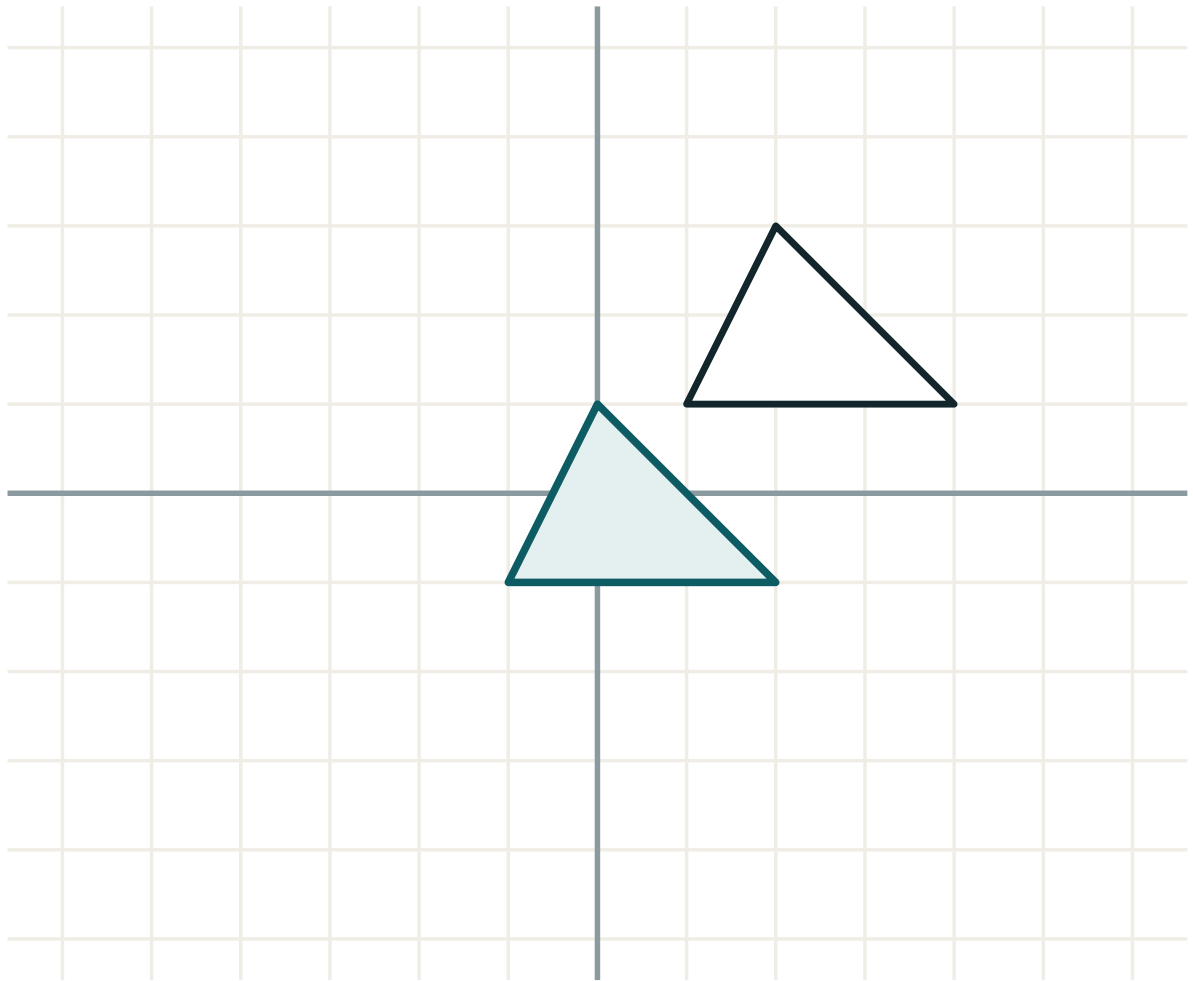
ANSWER

36 — the whole base

03 Interactive model

Shine a torch straight down on a tilted card and measure the shadow.

A projection



transformation **translation**

lengths **unchanged**

angles **unchanged**

orientation **kept**

04 Terminology

Write these on the board at the start. Learners meet the national programme in Uzbek or Russian and the Cambridge papers in English — the three columns are the same idea.

ENGLISH	O'ZBEKCHA	РУССКИЙ
Orthogonal projection	Ortogonal proyeksiya	Ортогональная проекция
Plan (top view)	Yuqoridan ko'rinish	Вид сверху
Front elevation	Old ko'rinish	Вид спереди
Side elevation	Yon ko'rinish	Вид сбоку
Technical drawing	Chizmachilik	Черчение
Hidden line	Ko'rinmas chiziq	Невидимая линия
Projection of an area	Yuza proyeksiyasi	Проекция площади
Scale	Masshtab	Масштаб
Isometric drawing	Izometrik chizma	Изометрия

Stress the words in bold in the explanation above; ask for the Uzbek and Russian back before moving on.

05 Quick check**Check yourself**

Orthogonal projection has the direction:

at any angle

perpendicular to the plane

along the plane

at 45°

The plan of a solid is the view from:

the front

the side

above

below

Circle, rectangle, rectangle is:

a cone

a cylinder

a sphere

a cube

A figure of area S at φ projects to:

$$S \sin \varphi$$

$$S \cos \varphi$$

$$S \tan \varphi$$

$$S$$

A square of side 10 at 60° projects to area:

$$50$$

$$86.6$$

$$100$$

$$25$$

06 Practice · 21 problems

Seven at each level. Set the easy row for everyone, the medium row for the main body of the class, and the hard row for those who finish early.

Easy · warm the idea up

7 PROBLEMS

1. The plan is the view from

ANSWER above

2. The front elevation shows

ANSWER length and height

3. Hidden edges are drawn

ANSWER dashed

4. Circle, circle, circle is

ANSWER a sphere

5. Square, square, square is

ANSWER a cube

6. Circle, triangle, triangle is

ANSWER a cone

7. S' for $S = 20, \varphi = 60^\circ$

ANSWER 10

Medium · the standard question

7 PROBLEMS

1. Circle, rectangle, rectangle is

ANSWER an upright cylinder

2. Square, triangle, triangle is

ANSWER a square pyramid

3. Triangle of area 48 at 30° : its plan

ANSWER $24\sqrt{3}$

4. Square of side 10 at 45° : its plan area

ANSWER $50\sqrt{2} \approx 70.7$

5. A figure projects to half its area: the angle

ANSWER 60°

6. A figure projects to nothing: the angle

ANSWER 90°

7. Draw the three views of a cuboid $4 \times 3 \times 2$

ANSWER three rectangles: 4×3 , 4×2 , 3×2

Hard · stretch and reason

7 PROBLEMS

1. Cube edge 6: projection of the section ABC_1D_1 on the base

ANSWER 36

2. Cube edge a : projection of the hexagonal midpoint section on the base

ANSWER $\frac{3}{4}a^2$

3. A circle of radius 5 at 60° : its plan

ANSWER an ellipse of area 12.5π

4. A regular hexagon of side 4 at 45° : its plan area

ANSWER $12\sqrt{3} \times \frac{\sqrt{2}}{2} \approx 29.4$

5. Two different solids with the same three views: give an example

ANSWER A cube, and a cube with an internal cavity

6. Prove $S' = S \cos \varphi$ for a triangle with a side on the edge

ANSWER The height is scaled by $\cos \varphi$

7. A roof face of 60 m^2 at 35° : its plan area

ANSWER $\approx 49.1 \text{ m}^2$

07 Homework

Homework — 6 tasks

Draw the three views aligned, to one scale, with hidden edges dashed.

1. Draw the plan, front and side elevations of a cuboid $5 \times 3 \times 2$.
2. Draw the three views of a square-based pyramid and of an upright cylinder.
3. Name the solid with plan a circle, front a triangle and side a triangle.
4. A rectangle of area 72 cm^2 lies at 40° to the horizontal. Find the area of its plan.
5. Cube of edge 8: find the area of the projection of the section ABC_1D_1 on the base.

6. Explain in three sentences why an orthogonal projection can be measured but a general parallel projection cannot.